Unit 6 Actor-Critic methods - hybrid architecture combining value & policy based methods that helps to stabilize the training & reduce variance using:

- An actor that controls how our agent behaves (policy-based)
- A Critic that measures how good the action taken is (value-based)

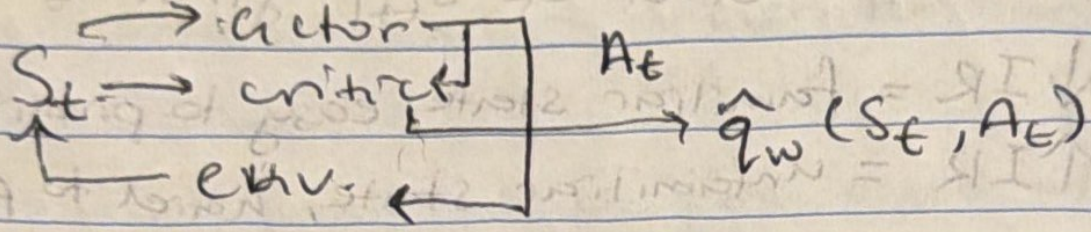
Stochasticity in the env. & in the policy lead to trajs. w/ dif. returns → high variance.

Variance can be reduced by using sampling from many traj. → this reduces sample efficiency

- Critic observes the actor's actions & provides feedback
  - ↳ a **policy** that controls how the agent acts $\pi_\theta(s)$
  - ↳ a **value** to assist the policy update by measuring how good the action taken was $\hat{q}_w(s,a)$

**Actor-Critic Process**

__Step 1__ - at timestep $t$, get state $S_t$, pass as input through Actor & Critic, policy outputs $A_t$



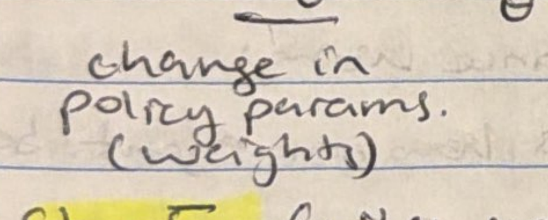

~~Critic~~ __Step 2__ - Critic computes how good it was to take that action at that time (Q-value) (basically rew?)

__Step 3__ performing action $A_t$ outputs $S_{t+1}$ & $R_{t+1}$

__Step 4__ - Actor updates the policy params. using Q-val

$$\underbrace{\Delta\theta}_{\text{change in policy params. (weights)}} = \alpha \nabla_\theta (\log \pi_\theta(s,a)) \underbrace{\hat{q}_w(s,a)}_{\text{action value estimate}}$$

__Step 5__ - Critic updates the val. params.




$$\Delta w = \beta(R(s,a) + \gamma \hat{q}_w(S_{t+1}, a_{t+1}) - \hat{q}_w(s_t, a_t)) \nabla_w \hat{q}_w(s_t, a_t)$$

Advantage in Actor-Critic (A2C)

- The advantage function calculates the relative advantage of an action compared to others taken in that state: how taken an action in a state compares to the average val. in that state.

Advantage func.

$$\underbrace{A(s,a) = Q(s,a)}_{\text{q-val for that action}} - V(s)$$

V(s) — average value of the state.

$A(s,a) > 0$ - pushed in that direction

$A(s,a) < 0$ - pushed away from that direction.

$$A(s,a) = Q(s,a) - V(s)$$

$$= \underbrace{r + \gamma V(s') - V(s)}_{\text{TD error}}$$